

Abbas Taqvi

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SUMMARY

Motivated **Data Engineer** with an experience in **AI/ML engineering** and a strong background in **Big-Data management, ETL, Query Engines, DataLakes, Warehouses** and **Lakehouses**. Well-versed in **Deep-Learning, LLMs and Natural Language processing**. Demonstrated research and collaboration abilities. Looking for a role with a room to take ownership and grow.

TECHNICAL SKILLS

Programming Languages: Python, Scala, Rust, Golang, Javascript, SQL, NoSQL

Data Engineering: Apache Spark, Apache Airflow, Kafka, Trino/Presto/Athena, Iceberg, Hive/Hadoop, Databricks, AWS S3, Glue

AI Engineering: LangChain, LangGraph, Qdrant, Weaviate, TensorFlow, PyTorch, Keras

Clouds and Software: AWS, Azure, Google CP, Linux (long-time desktop user), Docker, Kubernetes, DataGrip, AWS Lake Formation

EXPERIENCE

Data Engineer I

Logiciel Services LLC

May 2024 – Nov 2025

United States – Remote

- I was recruited as a **Founding Engineer** in the Data and AI department, **Designed and developed the data-infrastructure** including Spark and Kafka Jobs, Query Engines, Orchestrators, Data-Lakes, Data-Mesh and Data-Warehouses through state-of-the-art Lake-house architectures.
- Developed Lakes with financial data ranging to **30B records** that can be **queried within a second** with adequate where clauses.
- Developed a **domain-specific Text2SQL chatbot** for querying financial logs from Lakes, that'd allow business teams and stakeholders at **Velocity Capital** to query without needing a developers to write SQL, smoothening process by 41%.
- Developed a **Rust framework** for **ETL** (Ferroflow) for EOD processing of FIX logs along with Nifi pipeline, speeding up the legacy .NET utility process by **10x, saving 400 minutes daily** of AWS EC2 instance

Junior Data Engineer

Improdata

Jan 2021 – May 2022

Pinedale, Wyoming, US – Remote

- Worked on Azure Databricks (PySpark) for risk management of cyber-security data, building bronze, silver and gold layers.
- Worked on automated pipeline development using GCP Cloud Functions to transfer compressed json bulk data from local to GCS and population to BigQuery

DevOps Intern

Pullflow Inc.

July 2023 – Aug 2023

San Jose, California, US – Remote

- Worked on AWS EKS (Kubernetes), MicroK8s and Docker. Was responsible for designing and implementing the infrastructure that was on AWS to on-premise server.
- Worked on Budibase Dashboarding and hosting it on pullflow subdomain using traefik for loadbalancing.

Data Engineering Intern

United Bank Limited

May 2023 – Jun 2023

Karachi, Sindh, Pakistan – Onsite

- Built an Informatica pipeline for streaming UBL's data from Qatar and Bahrain into Hive Data Lake.
- Participated in architecture analysis to ensure production AI pipelines on UBL Pakistan's data work for Qatar and Bahrain.

EDUCATION

Università degli Studi di Milano-Bicocca

M.S. in Data Science

Milan, Italy

Oct 2025 – Present

Habib University

B.S. in Computer Science

Karachi, Pakistan

Aug 2020 – May 2024

PROJECTS

Kuza: FIX Datalake

Jan 2025 – Present

Logiciel Services LLC

Spark, Airflow, Kafka, Trino, AWS, PostgreSQL

- Implemented FIX ingestion pipelines for 375+ daily sessions across multiple environments.
- Developed Scala-based Spark workflows orchestrated using Airflow dynamic DAGs for scalable ingestion.
- Built an S3-based data lake with Trino + AWS Glue to enable unified schema management and fast querying.
- Improved operational reliability with schema control, file compaction logic and structured partitioning.

Ferroflow: ETL Framework

Jun 2024 – Jul 2024

Logiciel Services LLC

Rust

- Architected a Rust-based ETL framework inspired by NiFi, with reusable processors and composable pipelines.
- Achieved a 10× performance boost, reducing FIX-log processing from 500 minutes to nearly 50 minutes.
- Implemented optimized chunking, streaming and memory-mapped file strategies for large-file processing.
- Integrated Ferroflow in hybrid mode with NiFi to support smooth migration from legacy .NET workflows.

Metosi: FIX Logs Chatbot

Sept 2024 – Jan 2025

Logiciel Services LLC

Python, LangChain, LangGraph, SQL, Trino

- Built a multi-agent chatbot fine-tuned on FIX logs, trade events and FinTech operational concepts.
- Connected the chatbot to Trino for real-time SQL queries on the FIX datalake, enabling direct analytics.
- Reduced manual investigation time for business teams by around 40% through automated insights.
- Added retrieval, reasoning and execution flows using LangGraph for deterministic and interpretable outputs.

Astroflow: Bayesian Flow Networks (FYP)

Aug 2023 – May 2024

Habib University

PyTorch, Django, PostgreSQL

- Developed a Bayesian inference framework using normalizing flows for binary black-hole population modeling.
- Designed neural summarizers and feature extractors for unseen gravitational-wave-like data.
- Built a Django interface to run inference pipelines, organize experiments and visualize posterior samples.
- Optimized PyTorch training loops, enabling faster posterior sampling and improved model stability.

Yohsin: High-Performance Serialization Tool

Jan 2025

Hackathon Project

Rust, tokio

- Created a zero-copy, multi-threaded serializer/deserializer handling ~1M records (2.25GB) per second.
- Added support for random-access reads, range deserialization and state checkpointing.
- Used memory-mapped files and async I/O to greatly reduce RAM overhead while boosting throughput.
- Built benchmark suites to evaluate performance across varying record sizes and workloads.

INTERNATIONAL TEST SCORES

- International English Language Testing System (IELTS): **8.0 bands**
- Graduate Record Examinations - Maths (GRE): **165/170**

CERTIFICATIONS AND ACHIEVEMENTS

- **HU-TOPS Scholarship: Fully-funded Merit Scholarship** for Bachelor's in Computer Science
- **ICPC Regionals Asia Topi 2022: 19th position**
- **Winner – Game of Codes 5.0** Competitive Programming Competition

Relevant Certifications:

- Fundamentals of Accelerated Computing with CUDA C/C - **NVIDIA Deep Learning Platform**
- Building Data Lakes on AWS – **Amazon (Coursera)**
- ETL and Data Pipelines with Shell, Airflow, Kafka – **IBM (Coursera)**
- Spark Essentials with Scala - **RockTheJVM**